# From Owned Virtual Appliances to Scored Container Replicas: Configuration Parity, End-of-Life Package Repair, and Skill-Guided Conversion of Cybersecurity Laboratory Hosts

ZeroCrowd-Labs Project Team

*ZeroCrowd-Labs open research artifact repository*
*Corresponding author to be named on submission.*

**Abstract**

Cybersecurity laboratories depend on vulnerable virtual machines (VMs), yet a full guest—its own kernel, disk, and reserved memory—is an expensive unit to store, start, and reproduce. This paper asks whether an owned virtual appliance can be replaced by a Docker Compose replica that exposes the same laboratory host, and how completeness of such a replacement should be judged. We contribute (i) a six-step, skill-guided conversion procedure with explicit precondition and stop conditions for importing, offline administration, end-of-life package repair, inventory, replica construction, and parity scoring; (ii) an eleven-row configuration-parity rubric (listening ports, unpublished loopback services, package versions, users, sudoers, virtual hosts, web paths, databases, cron, mail, and application URL) scored as pass, partial, or missing, with a mandatory register of structural gaps that may never be promoted to passes; and (iii) a validated evaluation over eight laboratory replicas (88 rubric rows). The replicas pass 73 rows (83.0%; 95% Wilson interval [73.8, 89.4]), partially satisfy 15, and miss none, while their configured memory caps total 5,248 MB against 8,168 MB of guest RAM (35.7% lower; caps are configuration, not measured residency). Package versions (partial in six of eight labs) and listening ports (four of eight) dominate the partial rows, for reasons we trace to archive-revision drift, non-Debian guests, and nested container topologies. In a within-case longitudinal comparison, codifying the conversion rules raised one replica from 5 pass / 6 partial to 11 pass / 0 partial. A controlled file-lookup study (23 rule sentences, five repetitions) confirms that the decision rules reside in the skill documents (23/23 sentences) and are not recoverable from finished build files (7/184 lab-file hits). We report a data-validation audit that corrects three inconsistencies in the source record, and we delimit every claim: no second-agent timing, no measured memory residency, and no kernel equivalence is claimed.

*Keywords:* cybersecurity laboratories, cyber range, containers, virtual machines, configuration parity, reproducibility, end-of-life packages, LLM agents

## 1. Introduction

### *1.1. Problem statement*

Hands-on cybersecurity education and research rely on laboratories in which a learner or evaluator interacts with a deliberately configured host: published services invite enumeration, loopback-only services reward foothold thinking, and accounts, privilege rules, scheduled jobs, and mail complete the system under study (Du et al., 2007; Thompson & Irvine, 2018; Yamin et al., 2020). Community appliances distributed as virtual machine (VM) images—VulnHub-style "boot-to-root" boxes are the canonical example—package that host as a complete guest: its own kernel, a multi-gigabyte disk, and a reserved block of RAM whether or not the services are busy. Cyber-range research has repeatedly identified the cost of provisioning, resetting, and scaling such guests as a limiting factor, and container-based ranges have answered with lighter, faster-to-deploy scenarios (Chouliaras et al., 2023; Nakata & Otsuka, 2021; Ukwandu et al., 2020). Containers, however, change the object being delivered. A container shares the host kernel, replaces the guest init system with a supervisor or a single process,

and is usually rebuilt from current packages rather than from the guest's historical ones (Boettiger, 2015; Sultan et al., 2019). The laboratory may still *answer* on the expected port while silently ceasing to be the same host: a database engine has been substituted, a stored application URL has been rewritten, a loopback mail listener has vanished, or package versions have drifted to whatever the archive serves today.

This creates a measurement problem before it creates an engineering one. "The page loads" is not a fidelity criterion. What is needed is an explicit, checkable definition of *sameness* between an owned appliance and its container replica, a conversion procedure that can reach that definition repeatedly, and an honest account of what can never be made the same. Those three needs—a rubric, a procedure, and a gap discipline—motivate the present work.

A second difficulty is procedural scale. Converting one old appliance is a long chain of preconditioned operations: importing an appliance archive (OVA) without disturbing other guests, recovering local administration from a powered-off disk clone without attacking the running machine, repairing package management on an end-of-life (EOL) distribution whose suites have left the active mirrors, inventorying the guest without recording secrets, building a replica that keeps loopback services unpublished, and scoring the result against the inventory rather than against the build recipe. Each step contains traps that *look* successful locally—a newer base image that installs cleanly, a database TCP probe that succeeds while the application uses a different Unix socket, a supervisor that reports a mail daemon "stopped" during its first minute of startup. When the work is performed or assisted by a language-model (LM) agent, an additional risk appears: the reasoning that avoided a trap in one session does not automatically survive into the next conversion (Yao et al., 2023). The finished Dockerfile records the settings that were chosen; it does not record the conditions under which a different setting would have been a failure.

### *1.2. Contributions*

This paper makes three contributions.

**C1—A skill-guided conversion procedure (Section 2).** We formalise the conversion as six persistent procedural documents (*skills*), applied in precondition order: (1) import onto an existing NAT network, (2) offline administrator recovery on a disk clone, (3) EOL package-archive repair under stated trust limits, (4) secret-redacting guest inventory, (5) pinned replica construction, and (6) eleven-row parity scoring. Each skill states when it applies, what must already be true, the commands that count, the verification that closes the step, and the conditions under which the agent must stop. Traps discovered during scoring are written back into the skills, so the procedure—not a single session's transcript—is the durable artifact, in the checklist tradition studied by Haynes et al. (2009).

**C2—A configuration-parity rubric with a non-promotable gap register (Sections 2–3).** Completeness is scored on eleven rows—listening ports; unpublished loopback services; package versions; users; sudoers; virtual hosts; web paths; databases; cron; mail; and the application URL—each marked *pass*, *partial* (present with a stated substitution), or *missing* (absent, or a loopback-only service published). Three facts are kept in an explicit gap register even when every row passes: archive or snapshot package sources were required, the container shares the host kernel, and a process supervisor replaces the guest init system. A report that prints a password, hash, or flag value fails irrespective of its row scores. The rubric operationalises, for VM-to-container conversion, the outcome discipline that Jimenez et al. (2024) argue software-agent evaluation requires: success is a completed check, not a plausible narrative.

**C3—A validated, multi-laboratory evaluation with an audit trail (Section 4).** We apply the procedure and rubric to eight finished replicas spanning Debian 8–10, Ubuntu 14.04–18.04, CentOS 7, a Slackware-style live system, OpenBSD 5.9, single-container labs, a two-guest lab, and a nine-container lab (88 rubric rows). We quantify fidelity, configured-memory reduction, the anatomy of partial rows, and—in a single-case longitudinal study—the effect of codifying the rules (5 pass/6 partial before codification; 11 pass/0 partial after). We complement the case evidence with a controlled lookup study showing where the decision rules textually reside. Crucially, every number in the paper was re-derived from the primary artifact files by an automated validation script; three inconsistencies in the narrative record (most materially, a ninth replica scored in the files but described as "not converted" in the project README) are corrected and handled by

a sensitivity analysis rather than silently inherited (Section 4.1).

### *1.3. Related work*

*Cyber ranges and hands-on laboratories..* Yamin et al. (2020), writing in this journal, taxonomise cyber ranges by architecture, scenarios, functions, and tools, and Ukwandu et al. (2020) review range and testbed trends toward lighter, more composable infrastructure. The SEED project established the value of standardised, reproducible hands-on security exercises (Du et al., 2007); Labtainers showed that Docker containers can deliver consistent, individually parameterised exercises on modest student hardware with automated assessment (Thompson & Irvine, 2018); and SecGen addressed the static nature of vulnerable VMs by generating randomised vulnerable scenarios (Schreuders et al., 2017). Container-native ranges go further: UNIWA's autonomous platform builds complex topologies from infrastructure-as-code on container services (Chouliaras et al., 2023), while CyExec* reports near-perfect vulnerability reproducibility with roughly half the memory and a fraction of the storage of VM scenarios (Nakata & Otsuka, 2021). That literature optimises scenario *delivery*. It does not define when a container rebuilt from a *specific owned appliance* may be said to reproduce that appliance's listeners, accounts, package revisions, and stored application state—the fidelity question studied here. Our replicas are also not redistributed vulnerable images: they are local replicas of owned appliances, built and scored under the containment rules of Section 3.

*Containers versus virtual machines..* Felter et al. (2015) found containers matching or exceeding VMs on most performance measures, Bernstein (2014) and Merkel (2014) describe the packaging model that makes containers cheap to build and start, and Naik (2016) simulates operational gains from migrating VM deployments to Docker. Sultan et al. (2019) catalogue the security price of the shared-kernel model. Those results motivate replacing a guest with a container and simultaneously explain why the replacement cannot be total: kernel, init, and device behaviour are structurally different. Our gap register is the reporting device that keeps that difference visible after the port numbers start answering.

*Reproducible environments and package archaeology..* Boettiger (2015) frames Docker as an answer to computational irreproducibility; Nüst et al. (2020) distil Dockerfile practice, including stated base images and pinned versions; Lamb & Zacchiroli (2022) treat reproducible builds as a supply-chain integrity check; and Caneill et al. (2017) document two decades of Debian history that current mirrors no longer serve. The conversion problem adds an agent-facing question none of those papers answers: when an EOL suite survives only on an archive or dated snapshot, under exactly what conditions may an agent disable validity-window checking, add `trusted=yes`, or pin a snapshot—and how must that repair be reported so it is not mistaken for fidelity? Section 2 states those rules and Section 4 scores their consequences.

*Infrastructure-as-code verification..* Guerriero et al. (2019) report that practitioners lack practical tests for infrastructure code; Hummer et al. (2013) test configuration automations for idempotence; and Shambaugh et al. (2016) verify Puppet configuration determinacy with a solver. The parity rubric is a weaker, empirical relative: it compares a running replica against a redacted guest inventory, row by row, from live evidence (`ss`, `dpkg-query`/`rpm -q`, HTTP status, `visudo -c`), and it fails closed—a non-running replica scores its runtime rows missing rather than inheriting the Dockerfile's intentions.

*Agents, checklists, and outcome-based scoring..* Yao et al. (2023) interleave reasoning and tool use, but the plan is generated inside the episode; a skill inverts that arrangement by making the procedure persistent and prior to the episode. Haynes et al. (2009) show—in a clinical setting whose effect sizes we do *not* transfer—that checklists change work of this shape by making required actions and pause points explicit. Jimenez et al. (2024) score LM software agents by completed issue resolution (their strongest evaluated model resolved 1.96% of 2,294 real issues); we adopt the outcome discipline, not the benchmark: our check is parity, and a shared kernel may not be scored as reproduced. No prior work we are aware of states a VM-to-container parity rubric, the EOL-apt trust rules, or the supervisor/mail scoring traps for this conversion task. Table 1 positions the contribution against the closest literature.

Table 1: Positioning against related work. "—" means the work does not address the dimension. This paper is the only row that combines a VM-specific parity rubric, EOL-archive trust rules, and a non-promotable kernel/init gap register.

| Work | Container labs | Fidelity rubric | EOL rules | Gap register | Procedure |
|---|---|---|---|---|---|
| Yamin et al. (2020); Ukwandu et al. (2020) | Survey | — | — | — | — |
| SEED (Du et al., 2007); Labtainers (Thompson & Irvine, 2018) | Yes | — | — | — | — |
| SecGen (Schreuders et al., 2017) | VM generation | — | — | — | — |
| UNIWA CR (Chouliaras et al., 2023); CyExec* (Nakata & Otsuka, 2021) | Yes | Scenario-level | — | Kernel limits noted | — |
| Felter et al. (2015); Naik (2016); Sultan et al. (2019) | Performance/ security | — | — | Kernel analysed | — |
| Boettiger (2015); Nüst et al. (2020); Lamb & Zacchiroli (2022) | Packaging | Version pinning | Historical suites | — | — |
| Guerriero et al. (2019); Hummer et al. (2013); Shambaugh et al. (2016) | IaC testing | Config. checks | — | — | — |
| ReAct (Yao et al., 2023); SWE-bench (Jimenez et al., 2024) | — | Task success | — | — | In-episode |
| **This paper** | **Replica of owned VM** | **11-row parity** | **Archive/ snapshot rules** | **Kernel, init, archive** | **Persistent skills** |

### *1.4. Scope, ethics, and paper organisation*

The study covers owned appliances converted and run locally for education and evaluation. It contributes no attack procedure: the parity report deliberately contains no exploit steps, credentials, hashes, or flag values, and inventorying stops at version recording where a check would require attacking a service. Student-facing variants rebuild the same topologies on a current Debian 12 base with `PermitRootLogin no` and learner-chosen passwords (Section 3). Section 2 defines the method (research questions, procedure, rubric, deployment model, and evidence discipline). Section 3 describes the implementation mechanics. Section 4 presents the experimental design and results. Section 5 discusses interpretation, security implications, validity threats, and limitations. Section 6 concludes.

## 2. Methodology

### *2.1. Research questions and design overview*

The study is an artifact-centred case series with one embedded longitudinal case and one controlled text-lookup study. It follows case-study and empirical-reporting discipline for software and systems artifacts (Kitchenham et al., 2002; Runeson & Höst, 2009; Wohlin et al., 2012): research questions are fixed first, evidence sources are named before results, and every quantitative claim is traceable to a primary file.

- **RQ1 (Fidelity).** How completely do skill-guided container replicas reproduce their owned guest appliances under an eleven-row configuration-parity rubric, and where do partial rows concentrate?

- **RQ2 (Resource configuration).** How do configured replica memory caps compare with guest RAM allocations, and is any fidelity difference associated with the size of the configured saving?
- **RQ3 (Rule codification).** Within the one lab scored before and after the rules were codified (Raven), which rows changed, and what did the fixes consist of?
- **RQ4 (Rule residence).** Are the decision rules recoverable from the finished build files alone, or do they reside in the skill documents an agent is instructed to read?

RQ1 and RQ2 are answered descriptively over the eight-lab corpus (Section 4.2). No null-hypothesis test is offered for them: labs are heterogeneous cases, rubric rows within a lab are not independent trials, and $n = 8$ cannot support population inference. Two exploratory Spearman correlations are reported *as exploratory*, with their sample size attached. RQ3 is a single-case before/after comparison and is interpreted as such—it demonstrates that the codified rules address the observed failure modes, not that codification caused a measured agent speedup. RQ4 is answered by exact string lookup, a measurement of text residence; we are explicit that file-read milliseconds are not language-model latency and that no second agent was timed.

### *2.2. The six-skill conversion procedure*

Figure 1 shows the pipeline and Figure 2 the resulting deployment architecture. Table 2 states each skill's contract. The ordering is by precondition, not by convenience: inventory before build (the build consumes inventory facts, not guesses), archive repair before inventory where the guest must be booted to be inventoried, and scoring last and independently of the build recipe.

**Step 1—Import (`importing-ova-nat-network`).** Register exactly one appliance in VirtualBox; attach only that VM's enabled adapters to a pre-existing NAT network; record the VM list before and after. *Stop* if the NAT network is absent, the name collides, or any other VM changed. This containment rule is what makes a series of conversions composable on one host.

**Step 2—Offline recovery (`recovering-linux-admin-offline`; conditional).** Only when the guest password is unknown and the machine is powered off: clone the disk (RAW), mount the clone (handling LVM via device-mapper nodes where necessary), add a local administrator (uid $\geq 1000$, `sudo` on Debian/Ubuntu, `wheel` elsewhere) and set passwords inside the mount, verify by login (serial console where available), and convert back. The operator password is chosen at runtime and is never written to a skill, inventory, or report. *Stop* rather than escalating to an online attack; an encrypted (LUKS) root without a passphrase is a terminal stop.

**Step 3—EOL package repair (`repairing-eol-apt-archives`).** When `apt-get update` fails because the suite left the active mirrors: keep suite and component names, rewrite hosts to `archive.debian.org` / `old-releases.ubuntu.com` (CentOS guests use the vault mirror instead), disable the validity window (`Acquire::Check-Valid-Until "false"`), and add `trusted=yes` *only* to archived lines that still fail with `EXPKEYSIG`, expired signature, or `NO_PUBKEY`. If the old-releases host 404s while the active archive still serves the suite, move the lines back—trust is never extended to a current mirror. Verification is `apt-get update` and a simulated install (`apt-get install -s bash`) both exiting 0. The repair is legal and remains a registered gap (Section 2.3).

**Step 4—Inventory (`inventorying-linux-vm`).** Record, from the running guest where it can be booted and otherwise from unit files and configuration (the provenance is itself recorded): listeners with bind address and loopback status; human users with uid, home, and shell; sudoers (validated with `visudo -c`); service package version strings; virtual hosts, document roots, and web paths; the application database name, user, host, and socket; cron (root crontab, `cron.d`, user crontabs); and mail listeners. Passwords, hashes, private keys, and flag values are replaced by a redaction token at capture time.

**Step 5—Build (`building-docker-vm-replica`).** One Compose project per lab; base image from the guest's distribution codename where the guest system can be a Linux container base; *pin the inventoried package revisions* (dated snapshot where the archive no longer carries them) and do not call the package row a pass until `dpkg-query -W` (or `rpm -q`) matches the inventory strings; install the inventoried database engine, not a substitute (MariaDB for MySQL is a partial until

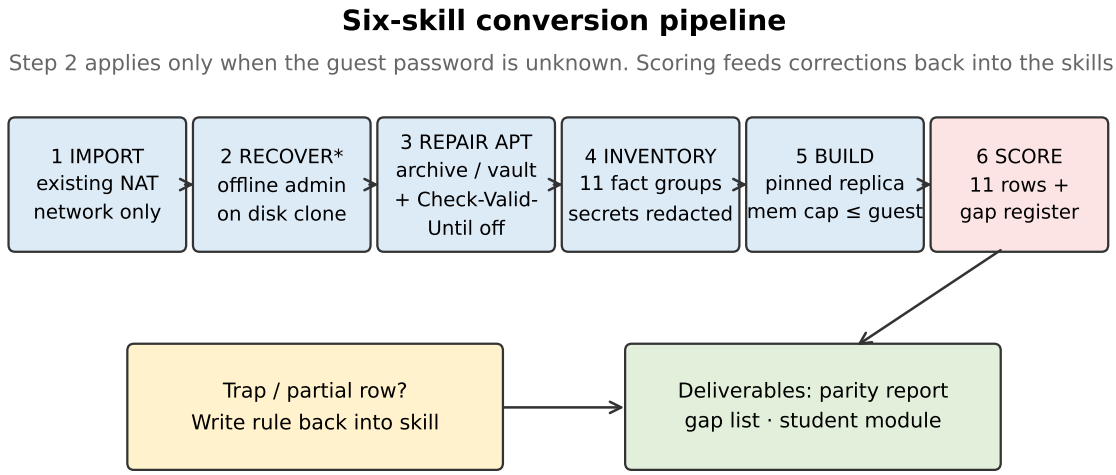



Figure 1: Six-skill conversion pipeline. Step 2 is conditional (unknown guest password). Scoring (Step 6) is performed from live evidence, and any trap or partial row it exposes is written back into the responsible skill before the replica is treated as finished.

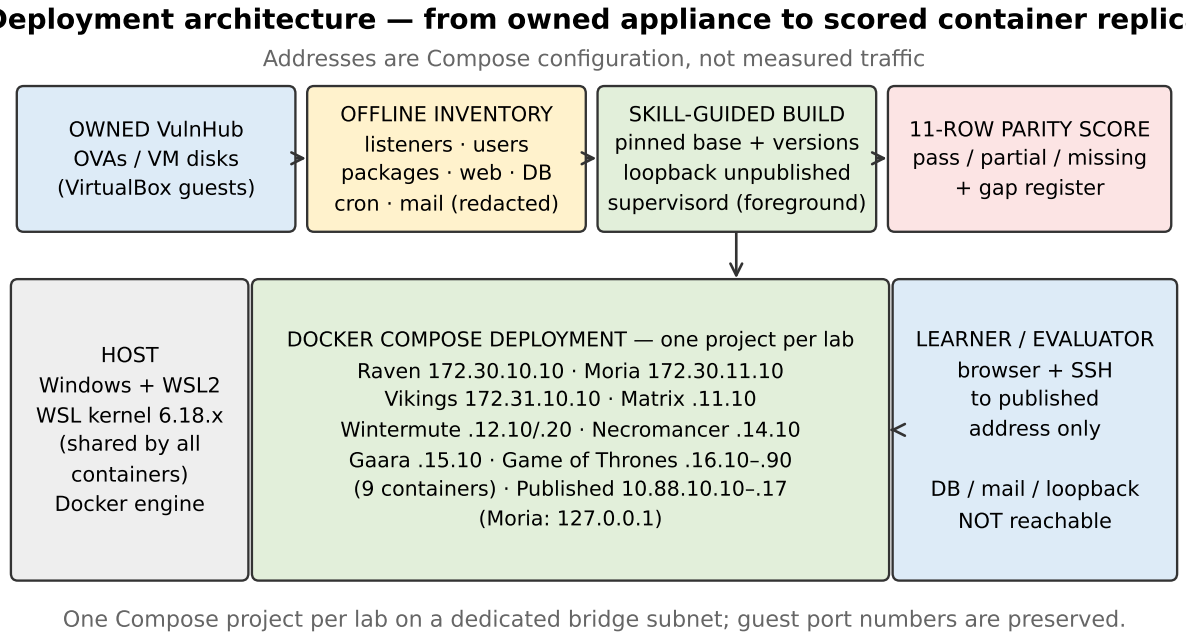



Figure 2: Deployment architecture. One Compose project per lab on a dedicated bridge subnet; publication only on a dedicated local address range (10.88.10.10–.17; Moria is the loopback-published exception). Container addresses (172.30/172.31.*) and published addresses are different layers of the same deployment and are both configuration, not measured traffic.

removed); publish exactly the guest's non-loopback port numbers, on the dedicated address where one is requested, leaving loopback services unpublished and never renumbering guest ports to dodge a busy local port; map the guest hostname to the published address instead of rewriting the stored application URL; initialise the application database once, reading its password from the configuration file as text and pointing the runtime at the socket the server actually creates; supervise long-running processes *in the foreground* (canonically `sendmail -bD`, not the forking `-bd`); and cap memory at or below the guest RAM allocation. Where the guest cannot be the base (OpenBSD, Slackware package format, a guest that itself ran a Docker engine), the substitution is chosen deliberately and scored partial where the rubric requires it.

**Step 6—Score (`evaluating-docker-vm-parity`).** Score the eleven rows from the inventory and from live commands against the running container—never from the Dockerfile alone. A non-running replica scores runtime rows missing. Mail is scored from listening sockets (`ss`) after the daemon's startup delay (about a minute for Sendmail 8.14), not from supervisor status in the first seconds. Archive apt alone does not make matching version strings partial. Publishing a loopback-only service is a missing row, not a convenience. The report must contain the gap register and no secrets, hashes, flags, or exploit steps. A seventh skill, `designing-labscenarios`, consumes inventories and topologies to draft *new* laboratory briefs (hosts, published vs. loopback services, roles; no attack path or credentials) and is used in teaching, not in conversion scoring.

### *2.3. The parity rubric and the gap register*

Table 3 defines the eleven rows. Three design choices deserve emphasis. First, the unit is the *row*, scored strictly: a package row passes only on string equality with the inventory (e.g., Apache `2.4.10-10+deb8u12` is not "close enough" to `2.4.25`); where several packages define a lab, the row is partial if the defining strings differ, even if some match. Second, *partial* is a positive reporting category, not a softened fail: it asserts the function exists with a named substitution, so readers can see exactly what was traded. Third, *missing* includes the security-relevant inversion—publishing a service the guest kept on loopback—because a replica that exposes a database the original kept private is not merely incomplete, it teaches the wrong topology.

Independently of the rows, every report carries a **gap register**: (i) version substitutions (empty when the package row passes); (ii) archive/snapshot/vault package sources, where used; (iii) second kernel—always "no": the container uses the host (WSL2) kernel, never the guest kernel; and (iv) init substitution (supervisord or a shell entrypoint in place of systemd/sysv/OpenBSD `rc`), plus deployment deltas (unrecreated NICs, collapsed guest networks, unloaded packet filters, and the configured-not-measured memory cap). The register is the paper's central honesty device: it makes the structurally unreproducible parts of a container conversion impossible to launder into a pass.

### *2.4. Deployment and dataset model*

Each lab is an independent Compose project with a dedicated bridge subnet and static container addresses; learners and evaluators reach only the published address (Figure 2). The dataset for RQ1–RQ2 is the set of primary artifact files per lab: `vm-inventory.md` (guest facts and provenance), `replica-parity.md` (row scores, substitutions, gap register), `docker-compose.yml/replica/compose.yaml` (caps, addresses, bindings), the Dockerfile(s) (base image and pins), and, where produced, `experiment.md` (lookup study). The student-facing `Educational-Module.md` files are derived teaching artifacts and are evidence only for the teaching discussion. Section 4.1 gives the dataset table and the audit that validated it.

### *2.5. Evidence discipline*

Three rules govern all reporting. (1) *Configuration is not measurement*: memory figures are Compose `mem_limit` values and VirtualBox allocations, not resident-set measurements; addresses are Compose configuration, not traffic captures. (2) *Provenance is data*: inventories collected from a running guest over a serial console are stronger evidence than inventories reconstructed from unit files of a guest that could not be booted, and tables carry that grade. (3) *Absence is reported*: where the record is silent (a version string not captured in an early round, a guest never booted), the cell stays empty and the text says so, rather than being reconstructed.

Table 2: Skill contracts: precondition, stop condition, and closing verification. The seventh (scenario-design) skill is not part of the conversion chain and is therefore listed separately in the text.

| Step | Skill | Stop condition (extract) | Verification that closes the step |
|---|---|---|---|
| 1 | importing-ova-nat-network | NAT network missing; name collision; another VM changed | VM list before/after differs only by the new guest |
| 2 | recovering-linux-admin-offline | Password known; next step would be an online attack; encrypted root, no passphrase | Administrator login verified on the clone |
| 3 | repairing-eol-apt-archives | `apt-get update` already exits 0; mirror still current; rewritten URL still 404s | `apt-get update` and simulated install exit 0 |
| 4 | inventorying-linux-vm | A secret, hash, or flag value would be recorded | Eleven fact groups recorded, secrets redacted, provenance stated |
| 5 | building-docker-vm-replica | Database host non-local (single-container pattern); loopback service would be published | Image present; container running; `docker inspect` confirms cap, address, bindings |
| 6 | evaluating-docker-vm-parity | Inventory missing; replica state unknown; check would require attacking a service | 11 rows scored pass/partial/missing; gap register completed |

Table 3: Parity rubric. Pass requires both configuration equality and the practical result (listener present, URL answering HTTP 200, rule parsing).

| Row | Pass criterion (summary) |
|---|---|
| Listening ports | Same port numbers/process roles listening; live evidence preferred |
| Loopback unpublished | Every guest loopback listener absent from host publication |
| Package versions | Inventoried version strings equal (`dpkg-query`/`rpm -q`) |
| Users | Same names, uids, homes, shells (added operator disclosed) |
| Sudoers | Same rules; `visudo -c` parses |
| Virtual hosts | Same vhost set, names, document roots |
| Web paths | Same paths present; expected HTTP status |
| Databases | Same engine and application schema (none/none passes) |
| Cron | Same crontabs and `cron.d` entries |
| Mail | Same mail package/listeners (none/none passes) |
| Application URL | Stored URL unchanged and answering |

## 3. Implementation—Mechanics

This section describes how the method is realised in the repository: the per-lab artifact layout, the build mechanics that make row passes achievable, the containment and teaching surfaces, and the failure traps that the skills now encode. It is deliberately mechanical: a reader should be able to repeat a conversion from this section and the skills without access to any conversion session.

### *3.1. Artifact layout*

Every lab occupies `labs/<name>/` containing the Compose entry point (`docker-compose.yml`, in most labs an `include` of `replica/compose.yaml`), the build context (`replica/` with Dockerfile, entrypoint, supervisor configuration, and service configuration), `vm-inventory.md`, `replica-parity.md`, `Educational-Module.md`, and the helper scripts produced while working the lab (disk mapping, mounting, download, and scoring scripts). Shared skills live in `skills/`; student-copied site files and images live in each lab's `resources/` and in the top-level `resources/`. Disk images, appliance archives, password files, and shadow extracts are intentionally not in the repository: the parity system must be auditable without the secrets it is forbidden to print. Keeping the inventory, the build, and the score as three separate files is what makes the without-skills condition of RQ4 well-defined—the build files are exactly what a later agent inherits if the skills are withheld.

### *3.2. Base images, pinning, and EOL repair in practice*

Replica bases follow the guest where a Linux container base exists: `debian:jessie` (Raven), `centos:7.3.1611` (Moria), `ubuntu:18.04` (Vikings), `debian:10` (Gaara), `debian:9` (Game of Thrones), `ubuntu:16.04` and `debian:stretch` (Wintermute's two containers), `ubuntu:14.04` (MrRobot, sensitivity corpus), and `python:2.7.14` for Matrix, whose guest is a Slackware-style Porteus live system whose one reproducible anchor

is the interpreter version (2.7.14) rather than a distribution. Necromancer's OpenBSD 5.9 guest cannot be a base on a Linux engine, so its replica is Debian 11 with nginx standing in for OpenBSD `httpd`; both substitutions sit in the gap register and the affected rows score partial.

Pinning is literal. Raven's Dockerfile, for example, writes six dated snapshot pairs plus the archive lines for Jessie, sets `Check-Valid-Until "false"` and `trusted=yes` on those archived lines only, and pins the inventoried revisions (Apache `2.4.10-10+deb8u12`, PHP `5.6.36+dfsg-0+deb8u1`, MySQL `5.5.60-0+deb8u1`, OpenSSH `1:6.7p1-5+deb8u4`, Sendmail `8.14.4-8+deb8u2`, cron, sudo) at priority 1001 before installing. Moria reaches the same string equality through the CentOS vault: its image is `centos:7.3.1611` and `rpm -q` returns the guest strings (httpd `2.4.6-45.el7.centos`, PHP `5.4.16-42.el7`, OpenSSH `6.6.1p1-33.el7_3`, postfix, vsftpd, chrony). Those two labs are the corpus's only package-row passes, and they demonstrate the rubric's asymmetry: archive use alone does not make a row partial, but the archive's *newer* security revision does. Five labs are partial precisely because the archive now serves a higher revision than the guest carried (e.g., Gaara's Apache `2.4.59-1~deb10u1` for `2.4.38-3+deb10u4`), which is a reproducibility fact about EOL ecosystems, not a build error that more effort at the same mirror could remove. Where the revision no longer exists on the archive, the build must name the snapshot timestamp that still carries it—the mechanism Raven's six snapshot lines encode.

Two build subtleties recur. Jessie-era apt can ignore `trusted=yes` and fail signature validation through the `gpgv` shim; the skill's remedy (a `Dir::Bin::gpg` stub emitting `GOODSIG/VALIDSIG` for the archived suite only, or `--force-yes` where that apt requires it) is scoped to the archived lines for exactly the trust reason stated in Step 3. And database initialisation is one-shot: a volume initialised by one engine may not be mounted into the other—in Raven, the MySQL 5.5 volume had to be created fresh after the MariaDB substitution was removed, a rule the build skill now states explicitly.

### *3.3. Supervision, sockets, and the mail trap*

Containers have no guest init, so entrypoints supervise the inventoried daemons in the foreground under supervisord (or, for Moria and single- process Game-of-Thrones containers, a shell entrypoint or the daemon's own foreground mode). Three mechanical rules in the skills come from observed mis-scores. First, *foreground means foreground*: `sendmail -bd` forks and exits, so the supervisor marks the program stopped while a child may still listen; `sendmail -bD` stays in the foreground and is the supervised form. Second, *score the socket, not the supervisor*: Sendmail 8.14 sleeps for roughly a minute after its startup DNS/syslog attempts, so ports 25/587 are scored from `ss` after that window; shortening resolver timeouts and providing `/dev/log` did not remove the sleep, which is why the rule is a wait, not a configuration change. Third, *the application's socket is the server's socket*: PHP's MySQL socket path is configured (`mysql-socket.ini` in both the Apache and CLI PHP configurations in Raven) to the socket the installed server creates (`/var/run/mysqld/mysqld.sock`); a successful TCP probe on 3306 does not evidence this, and an earlier build generation that trusted it scored the database row on the wrong signal. Equivalent small mechanics appear across the corpus: rpcbind plus foreground `rpc.statd` restores Raven's NFS listener row; Moria's chrony needs `CAP_SYS_TIME` in the container (`cap_add: SYS_TIME`) and its vsftpd is run IPv4-only with `background=NO`; Game of Thrones runs nine single-process containers (SSH, Apache, nginx×2, BIND, Dovecot, MariaDB, PostgreSQL, pure-ftpd) on one bridge because the guest itself was a Docker host—reproducing its topology, not its engine.

### *3.4. Networking and publication mechanics*

Publication is the security-critical mechanic. Compose binds only the inventoried non-loopback port numbers, and binds them to a dedicated local address (10.88.10.10–.17 in the primary corpus) rather than to all interfaces or to a contended 127.0.0.1. That choice preserves the guest's port numbers—a rubric requirement—without colliding with other replicas' loopback publications, and it keeps the stored application URL honest: Raven serves `http://raven.local/wordpress/` with the hostname mapped to the published address (`extra_hosts` / hosts entry) instead of rewriting WordPress's stored URL to a raw address, which an earlier generation did and which the rubric counts partial. Loopback services (MySQL on 3306, Sendmail on 25/587, Redis on 6379, SNMP on 161/udp, Tomcat's shutdown port,

Necromancer's entire HTTP/SNMP/Python set, Vikings' RPyC server on 18812) are present inside the container where the guest had them and absent from the Compose publication list; Docker's embedded resolver (127.0.0.11) is likewise never published. Moria is the documented exception, publishing 21/22/80 on 127.0.0.1, and Necromancer is the limiting case: only SSH is published and its application URL answers only inside the container, which the rubric scores as a pass because that *is* the guest topology. Guest networks that cannot be recreated (Wintermute's two host-only networks, Game of Thrones' eight isolated container networks) collapse to the lab bridge and are recorded as gaps.

### *3.5. Containment, secrets, and the teaching surface*

The conversion workflow is contained at each step: appliances attach only to an existing NAT network; offline administration edits a *clone*, never the registered disk, and never a running machine; inventory redacts at capture; parity reports fail if they contain secrets. The teaching surface is deliberately separated from the fidelity surface. Each lab's `Educational-Module.md` rebuilds the same topology—same listeners, accounts, and loopback decisions—on a current Debian 12 student VM (host-only 192.168.56.0/24 plus NAT for package access), with `PermitRootLogin no`, learner-chosen passwords excluded from hand-ins, and verification by `dpkg-query -W` lines and HTTP status codes. The modules therefore teach the inventoried system without distributing the historical vulnerable image, and their "what to hand in" (topology, accounts, package lines, verification statuses) mirrors the parity rows. A scenario-design skill drafts new briefs from existing inventories—setting, scale, published versus unpublished services, and a differentiating twist—explicitly without attack paths, flags, or passwords.

## 4. Evaluation

### *4.1. Experimental methodology*

*Units, variables, and scoring protocol..* For RQ1–RQ2 the unit of analysis is the parity row (11 per lab; 88 in the primary corpus), grouped by lab. The independent descriptors are the lab's guest system, topology (guests and containers), and inventory provenance; the dependent variables are the row outcome (pass/partial/missing), the strict pass count per lab, and the configured memory cap. Scoring follows Section 2.3 from live evidence recorded in `replica-parity.md`; two readings of the same file are the longitudinal data for RQ3. For RQ4 the unit is the rule sentence (23 sentences coded from the apt, build, and parity skills; listed in the repository as the experiment sentence set), searched exactly in two corpora—the three skill files (*with-skills*) and each lab's finished build/inventory/score files (*without-skills*; web trees, shadow extracts, and disk images excluded)—with five repetitions timed by `System.Diagnostics.Stopwatch` in one PowerShell process; the median is the third sorted value and repetition 1 is the cold read. All experiments were executed on 13 October 2026 in the project environment; the lookup script searched text, and no container was rebuilt and no second LM agent was run for this evaluation—a boundary we maintain in every claim.

*Dataset..* Table 4 summarises the primary corpus: eight labs, ten guest systems (Wintermute contributes two), thirteen replica containers, guest RAM from 512 to 2,048 MB, and five operating-system families. The corpus is a convenience case series—the owned appliances available to the project—not a sample of VulnHub; we make no prevalence claim beyond it. Inventory provenance is heterogeneous by necessity and is graded per lab: Raven, Moria, and Vikings were inventoried from booted guests (serial console in the first two), while Gaara, Matrix, Wintermute, Necromancer, and Game of Thrones could not be booted in the host environment (locked VirtualBox sessions, hung settings operations, or powered-off guests) and were inventoried from unit files, service configuration, startup scripts, and container metadata. Listener rows for the second group therefore rest on configuration evidence, and their parity reports say so; this is treated as a validity threat in Section 5, not smoothed over.

*Data validation audit (corrections applied)..* Before analysis we re-derived every headline number from the primary files with an automated audit script (retained with the paper materials; it re-parses all `replica-parity.md` tables, Compose caps, and experiment files and fails on mismatch). The audit confirmed the parity counts (73/15/0), the memory sums (8,168/5,248 MB), and the per-lab experiment

Table 4: Primary dataset (eight labs). RAM/cap are configured values (VirtualBox allocation; Compose `mem_limit`). Provenance grades the guest inventory: *live* = collected from the booted guest; *config* = reconstructed from unit/config files of an unbooted guest.

| Lab | Guest system (kernel) | Guest RAM | Cap | Containers | Published | Inventory |
|---|---|---|---|---|---|---|
| Raven | Debian 8.11 jessie (3.16.0-6) | 512 | 512 | 1 | 10.88.10.10 | live (serial) |
| Moria | CentOS 7.3.1611 (3.10.0-514.el7) | 1024 | 512 | 1 | 127.0.0.1 | live (serial) |
| Vikings | Ubuntu 18.04 bionic (4.15.0-154) | 1024 | 512 | 1 | 10.88.10.11 | live |
| Matrix | Porteus 4.0, Slackware-style (n/c) | 512 | 384 | 1 | 10.88.10.12 | config |
| Wintermute | Debian 9 (4.9.0-6) + Ubuntu 16.04 (4.4.0-116) | 2048 | 1024 | 2 | 10.88.10.13/.14 | config |
| Necromancer | OpenBSD 5.9 GENERIC | 512 | 384 | 1 | 10.88.10.15 | config |
| Gaara | Debian 10 buster (4.19.0-13) | 1024 | 512 | 1 | 10.88.10.16 | config |
| Game of Thrones | Debian 9 stretch (4.9.0-3) | 1512 | 1408 | 9 | 10.88.10.17 | config |
| Total (8 labs) | 10 guests, 5 OS families | 8168 | 5248 | 13 | — | 3 live / 5 config |

files. It also found three inconsistencies in the narrative record, which this paper corrects rather than inherits:

1. **The corpus is nine scored replicas, not eight.** The project README states that the MrRobot appliance "was not converted", has no Compose project, and is excluded from the scores. The repository contradicts that text: `labs/mrrobot-lab/` contains a Compose project, a built-and-running parity report (8 pass/3 partial/0 missing on Ubuntu 14.04), and a lookup experiment. We therefore define the *primary* corpus as the README's eight labs for comparability with the project's headline claims, and report a *sensitivity* corpus adding MrRobot (9 labs, 99 rows) wherever a headline number could change. The stale README sentence is flagged for correction in the repository.
2. **Two address layers were conflated in draft text.** Container bridge addresses (172.30/172.31.*), dedicated published addresses (10.88.10.*), and student host-only addresses (192.168.56.*) are three layers of three different deployments (replica, evaluator access, teaching rebuild). Tables in this paper label the layer they report.
3. **Three different lookup runs were not interchangeable.** The cross-lab lookup (Experiment A), the per-lab lookups (Experiment B, seven labs—Raven and Moria have no per-lab file), and an earlier Raven-only lookup preserved in draft text used different file sets and produced different medians (e.g., skills-corpus medians of 2.539 ms, 0.593–2.129 ms, and 1.014 ms respectively). They measure the same coverage fact on different corpora and are reported separately in Section 4.2.4; pooling them would be a measurement error.

### *4.2. Results*

#### *4.2.1. RQ1—Fidelity across eight replicas*

Across the primary corpus, replicas pass **73 of 88 rows (83.0%; 95% Wilson score interval [73.8, 89.4])**, partially satisfy 15 (17.0%), and miss none (Table 5, Figures 3–4). The interval is reported for scale only; rows are clustered within labs and are not Bernoulli trials. Two labs (Raven, Moria) pass all eleven rows; the median lab passes 9; the lowest, Game of Thrones, passes 7 with 4 partial. Adding MrRobot (sensitivity corpus) gives 81/99 pass (81.8%; [73.1, 88.2]), 18 partial, 0 missing—no conclusion in this paper changes direction under the sensitivity corpus.

The anatomy of the 15 partial rows is concentrated, not diffuse (Table 6, Figures 4 and 5). Four areas pass on every lab—unpublished loopback services, users, sudoers, and the application URL—which is to say the security topology and the identity surface are the most reproducible parts of these guests. Virtual hosts, web paths, databases, cron, and mail each fail strict pass once. The two systematic weaknesses are **package versions (partial in 6 of 8 labs; strict pass rate 25%)** and **listening ports (4 of 8; 50%)**, and their causes are classifiable (Table 7):

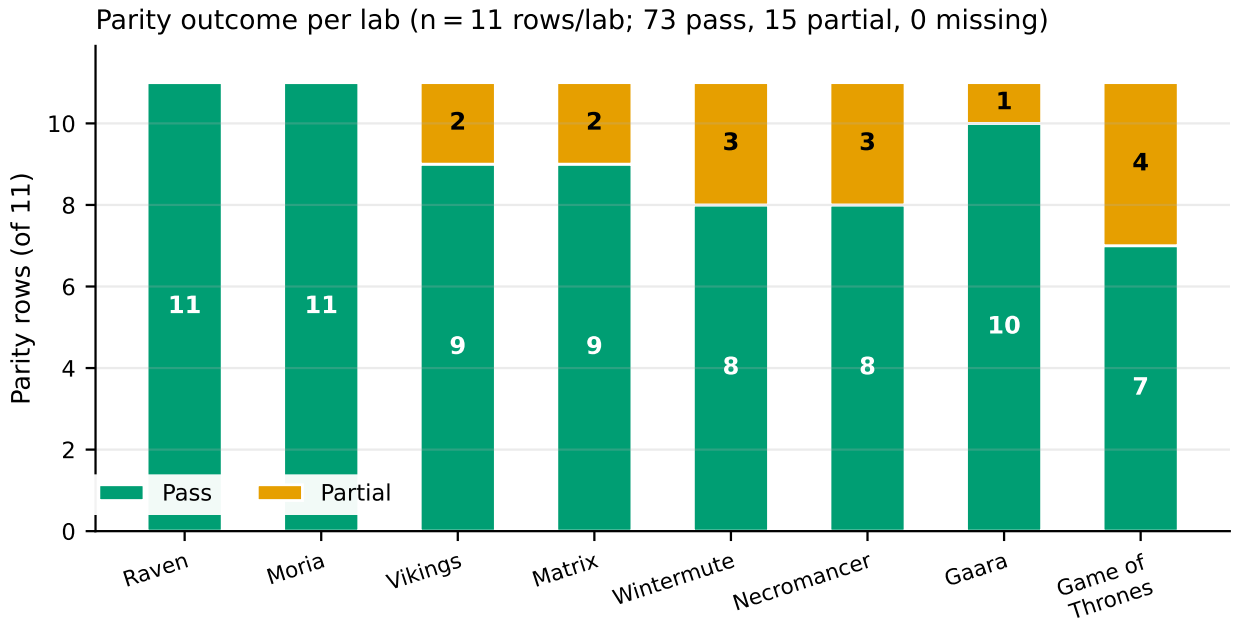



Figure 3: Pass/partial rows per lab ($n = 11$ rows per lab). No row in any lab is missing: every partial is a present function with a named substitution, not an absence.

Table 5: Parity outcome per lab (11 rows each). "Partial drivers" names the rows scored partial; full substitutions are in each lab's parity report and gap register.

| Lab | Pass | Part. | Miss. | Partial drivers |
|---|---|---|---|---|
| Raven | 11 | 0 | 0 | — |
| Moria | 11 | 0 | 0 | — |
| Gaara | 10 | 1 | 0 | packages |
| Vikings | 9 | 2 | 0 | ports, packages |
| Matrix | 9 | 2 | 0 | ports, packages |
| Wintermute | 8 | 3 | 0 | ports, packages, mail |
| Necromancer | 8 | 3 | 0 | packages, vhosts, cron |
| Game of Thrones | 7 | 4 | 0 | ports, packages, web, DB |
| **Total** | **73** | **15** | **0** | packages in 6/8 labs |

- *Archive-revision drift* (Gaara, Game of Thrones, Wintermute, and the archive component in Vikings): the EOL archive serves a newer security revision than the guest carried. The function is identical; the string is not, and the rubric scores strings.
- *Unreproducible guest substrate* (Matrix's Slackware packages on a Debian base; Necromancer's OpenBSD base, `httpd`, and periodic scripts on Debian 11): no Linux container base exists, so the row is partial by construction.
- *Guest-runtime collapse* (Vikings' systemd-resolved and containerd listeners; Game of Thrones' guest Docker engine and its nine images; Wintermute's un-recreated second guest services): listeners or engines that existed only as guest-runtime state are replaced by named stand-ins (e.g., nginx for the guest's port-10000 service, MariaDB for its MySQL image, Dovecot for its mail image) or recorded unbound until use (FTP passive range).
- *Single-string drift* (Wintermute mail: Postfix `3.1.15-0+deb9u1` for `3.1.8-0+deb9u1`): one defining string differs while the service is otherwise the guest's.

#### 4.2.2. *RQ2—Configured memory and its relation to fidelity*

Configured replica caps total **5,248 MB against 8,168 MB** of guest RAM—2,920 MB (35.7%) less

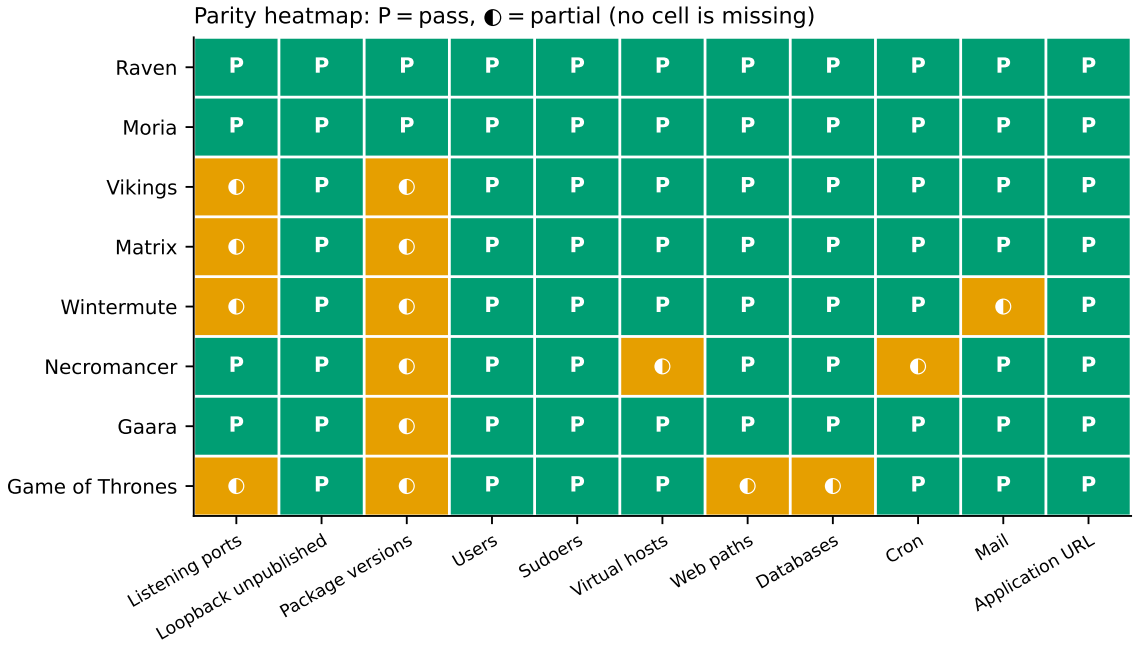



Figure 4: Parity heatmap (labs × areas). Partials cluster in the package-version column (six labs) and the listening-port column (four labs); identity, privilege, containment, and URL columns are uniformly pass.

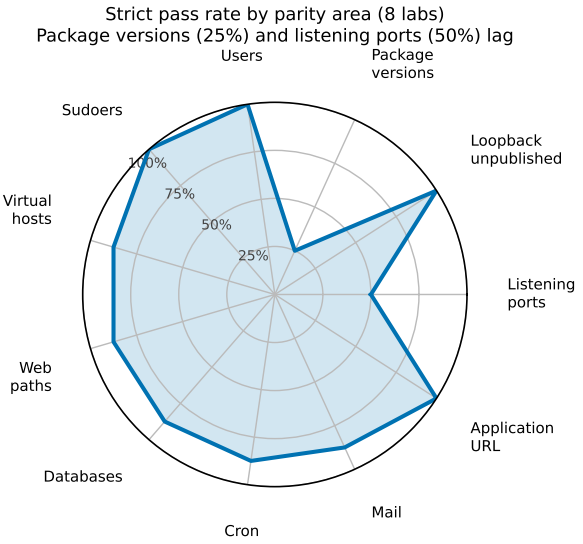



Figure 5: Spider (radar) view of strict pass rate by area. The two indentations—package versions (25%) and listening ports (50%)—are the corpus's fidelity frontier; all other areas sit at 87.5–100%.

Table 6: Strict pass by rubric area (8 labs). Users, sudoers, loopback containment, and the application URL are universally reproduced; package versions and listening ports are not.

| Area | Pass | Partial |
|---|---|---|
| Listening ports | 4 | 4 |
| Loopback unpublished | 8 | 0 |
| Package versions | 2 | 6 |
| Users | 8 | 0 |
| Sudoers | 8 | 0 |
| Virtual hosts | 7 | 1 |
| Web paths | 7 | 1 |
| Databases | 7 | 1 |
| Cron | 7 | 1 |
| Mail | 7 | 1 |
| Application URL | 8 | 0 |

configuration for the same published port surface (Figure 6). Per lab, caps are at or below the guest allocation by construction (the build skill forbids exceeding it): Raven is equal (512/512 MB, because Jessie-era services were sized to the guest), Moria, Vikings, Wintermute, and Gaara halve the allocation, Matrix and Necromancer retain 75%, and Game of Thrones retains 93.1% (1,408 of 1,512 MB, the sum of nine container caps). Guest RAM and cap are strongly rank-associated (Spearman $\rho = 0.87$, $n = 8$; Figure 7), which is

Table 7: Root-cause classification of all 15 partial rows (primary corpus). A row may carry one primary cause; counts sum to 15.

| Cause | Rows | Labs | Mechanism |
|---|---|---|---|
| Archive-revision drift | 5 | Gaara, Vikings, Wintermute (pkg), GoT, Wintermute (mail) | Archive serves newer EOL security revision than guest string |
| Unreproducible substrate | 3 | Matrix (pkg), Necromancer (pkg, vhosts) | Slackware/OpenBSD cannot be the Linux container base |
| Guest-runtime collapse | 4 | Vikings (ports), GoT (ports, web, DB) | Guest-runtime state replaced by stand-ins or uncopied trees |
| Live-evidence unavailable | 2 | Matrix (ports), Wintermute (ports) | Guest never booted; listener row rests on startup/config files |
| Substitution disclosed | 1 | Necromancer (cron) | Added job present; BSD periodic programs absent |

expected because the cap rule is defined relative to the guest; we report it as a design check, not a discovery. The substantive question—whether larger savings cost fidelity—shows no evident association in this corpus (saving vs. pass rows: $\rho = 0.14$, $n = 8$, exploratory; Figure 8b). Fidelity instead tracks topological complexity: replicas with more containers pass fewer rows ($\rho = -0.72$, $n = 8$, exploratory; Figure 8a), driven by Game of Thrones (9 containers, 7 passes) and Wintermute (2 guests, 8 passes) against five single-container labs at 8–11 passes. With eight heterogeneous cases both correlations are hypothesis-generating only. And the boundary of the claim matters: caps bound what the engine *may* allocate. No resident memory, startup time, or throughput was measured, so no operational saving is claimed—only that the laboratory's published surface is configured to run inside a smaller envelope, consistent with the direction (not the magnitude) of container/VM comparisons in the literature (Felter et al., 2015; Nakata & Otsuka, 2021).

#### *4.2.3. RQ3—What codifying the rules changed (Raven)*

Raven is the one lab scored twice on the same eleven rows: before the conversion rules were written into the skills (Round 1) and after the procedure was applied and the newly discovered traps were written back (Round 2). The replica went from **5 pass / 6 partial / 0 missing (45.5% strict pass) to 11 pass / 0 partial (100%)** (Figure 9). Table 8 names the six fixes. Each maps to a skill sentence that now exists independently of Raven: rebuild from the guest codename and pin the inventoried revisions; install the inventoried database engine on a fresh volume; install the package that owns a missing listener (`nfs-common` / `rpc.statd`); keep the distribution's PHP cron snippet; supervise Sendmail in the foreground and score its sockets after the startup minute; and preserve the stored application URL by mapping the hostname to the published address. Round 1 is instructive precisely because the container *worked*: it answered HTTP with Debian 9 packages and MariaDB 10.1 standing in for Debian 8 and MySQL 5.5. Under a "page loads" criterion the conversion was finished; under parity, six rows were partial. The structural gaps did not move: archive/snapshot apt with `trusted=yes` on the archived lines, the WSL host kernel (6.18.x) in place of the guest's `3.16.0-6-amd64`, and supervisord in place of sysv init remain in Raven's gap register in Round 2. That separation—rows improvable to pass, gaps permanently disclosed—is the rubric working as designed.

We state the inferential limit plainly. This is one case, scored by the project, with the rebuild and the rule-writing entangled: Round 2 differs from Round 1 both because the rules existed and because the replica was rebuilt under them. Rows within the case are not independent observations, and no statistical test is applied. The claim is therefore mechanistic, not causal-population: the six Round 1 failure modes are each addressed by a named rule, and the rescored artifact passes the rows those rules govern. Whether a *different* agent, given only the skills, reaches Round 2 on a *new*

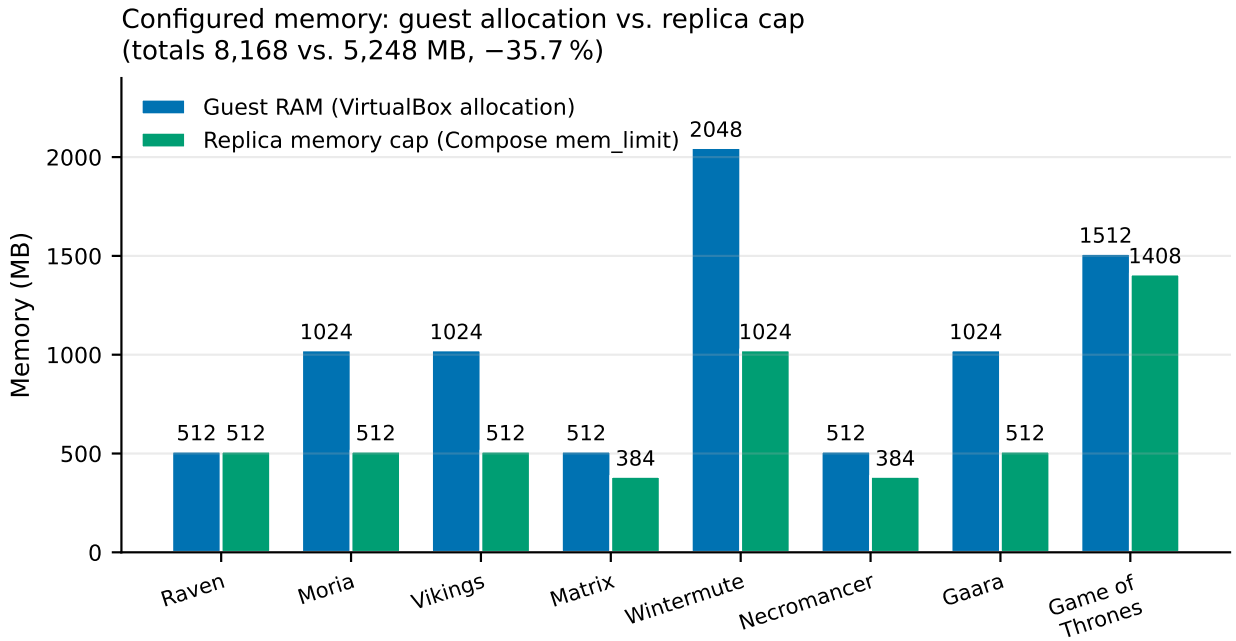



Figure 6: Configured memory by lab: VirtualBox guest allocation vs. Compose memory cap. Wintermute aggregates two guests (1,024 MB each) and two containers (512 MB each); Game of Thrones aggregates nine containers (256 + 6 × 128 + 2 × 192 MB).

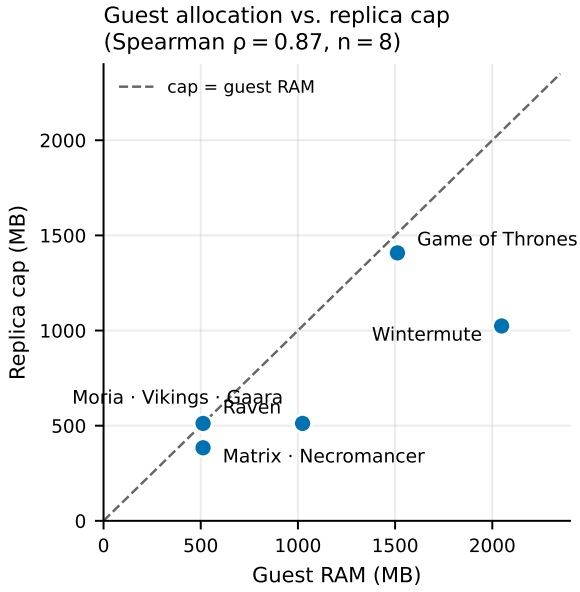



Figure 7: Guest allocation vs. replica cap with the identity line. Every lab sits on or below the line by rule; rank association ($\rho = 0.87$) reflects the capping rule, not an empirical law. Labs with identical (RAM, cap) coordinates are labelled jointly (Matrix + Necromancer; Moria + Vikings + Gaara).

appliance without assistance is the decisive test and has not been run (Section 5).

#### *4.2.4. RQ4—Where the rules reside*

Experiment A (cross-lab, 13 October 2026) searched the 23 coded rule sentences in the three skill files and, separately, in each primary lab's finished files (Table 9, Figure 10). The skills contain all 23 sentences (23/23). The eight lab corpora contain **7 hits in 184 opportunities (3.8%)**: seven labs contain exactly one sentence—the `Acquire::Check-Valid-Until "false"` assignment that necessarily survives into a Dockerfile or sources list—and Moria contains none, because its CentOS vault repair never writes the Debian assignment. In other words, 22 of the 23 decision rules (when `trusted=yes` is legal, which ports stay unpublished, why MariaDB-for-MySQL remains partial, why the kernel is never a pass, how long to wait before scoring mail) are textually absent from the artifacts a skills-less agent would inherit. Experiment B (per-lab files, seven labs) replicates the coverage result on independently assembled corpora: skills 23/23 in every lab's run; replica files 1/23 in every lab, again the `Check-Valid-Until` line.

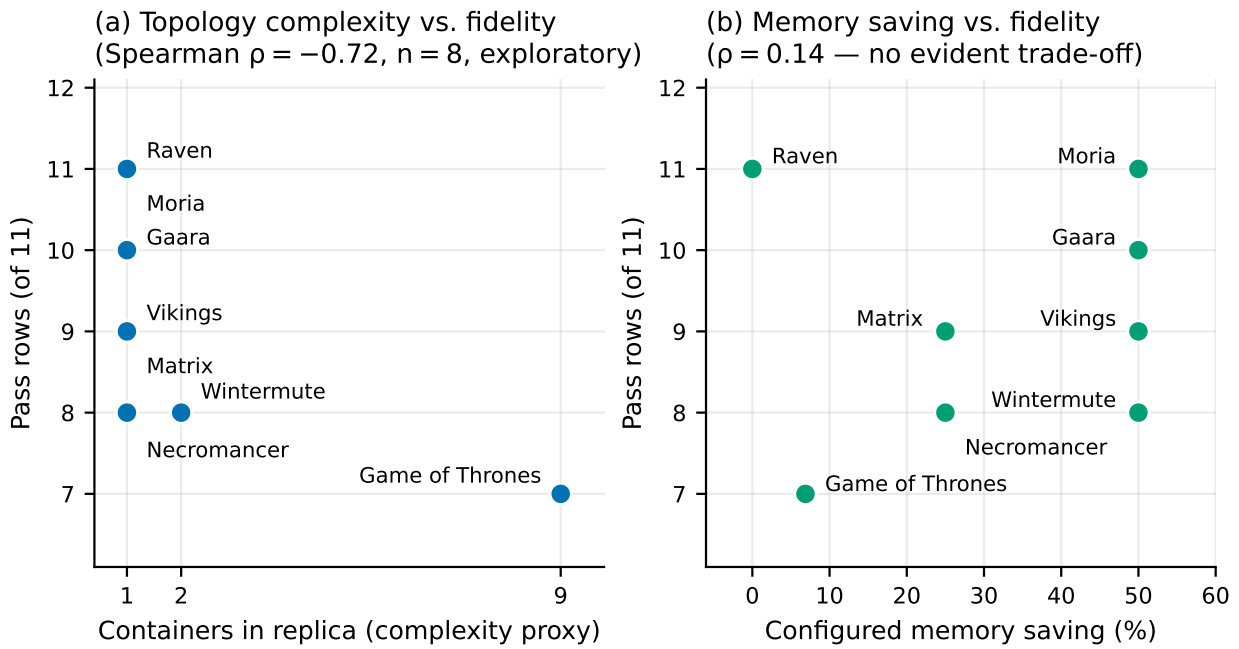



Figure 8: Exploratory associations ($n = 8$; hypothesis-generating only). (a) Fidelity falls as replica container count rises. (b) Fidelity shows no evident association with configured memory saving. Together they suggest complexity, not frugality, is the fidelity risk in this corpus.

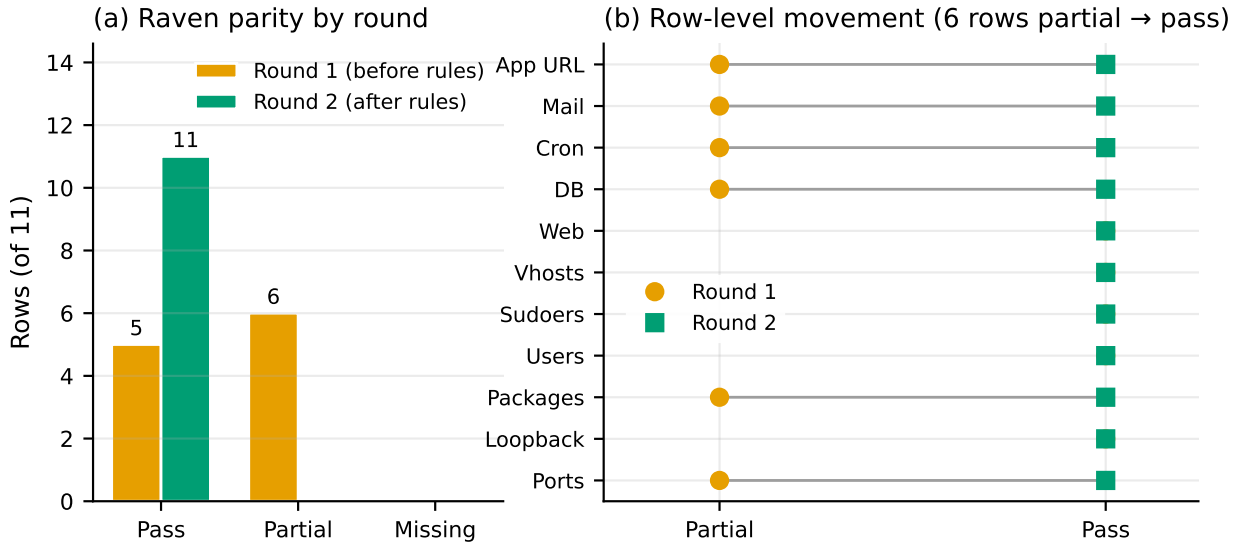



Figure 9: Raven longitudinal comparison ($n = 11$ rows). (a) Row counts by round. (b) Row-level dumbbell: six rows move partial→pass; five were already pass; none is missing in either round. The gap register (archive apt, shared kernel, supervisor) is unchanged and is not part of the 11 rows.

Table 8: The six Raven rows that changed between rounds, the codified fix, and the governing skill rule. Rows already passing (loopback, users, sudoers, vhosts, web paths) required no fix.

| Row (Round 1 → 2) | Fix applied | Rule now codified |
|---|---|---|
| Ports: partial → pass | Install `nfs-common`; run `rpcbind` + foreground `rpc.statd` | Score every inventoried listener from `ss`; a missing daemon is a build defect, not a note |
| Packages: partial → pass | Rebuild `FROM debian:jessie`; pin guest revisions via dated snapshots | Do not call the row a pass until `dpkg-query` matches inventory strings |
| Databases: partial → pass | Install `mysql-server` 5.5.60 on a fresh volume (MariaDB removed) | Install the inventoried engine; never mount one engine's data dir into the other |
| Cron: partial → pass | Keep the distro php5 `cron.d` snippet | A package-owned snippet absent from the replica is a partial row |
| Mail: partial → pass | Pin Sendmail 8.14.4; supervise `sendmail -bD`; score after ≈60 s | Score mail from sockets after the startup window, never from early supervisor status |
| App. URL: partial → pass | Keep `http://raven.local/wordpress/`; map the name to 10.88.10.10 | Map the hostname to the published address; do not rewrite the stored URL |

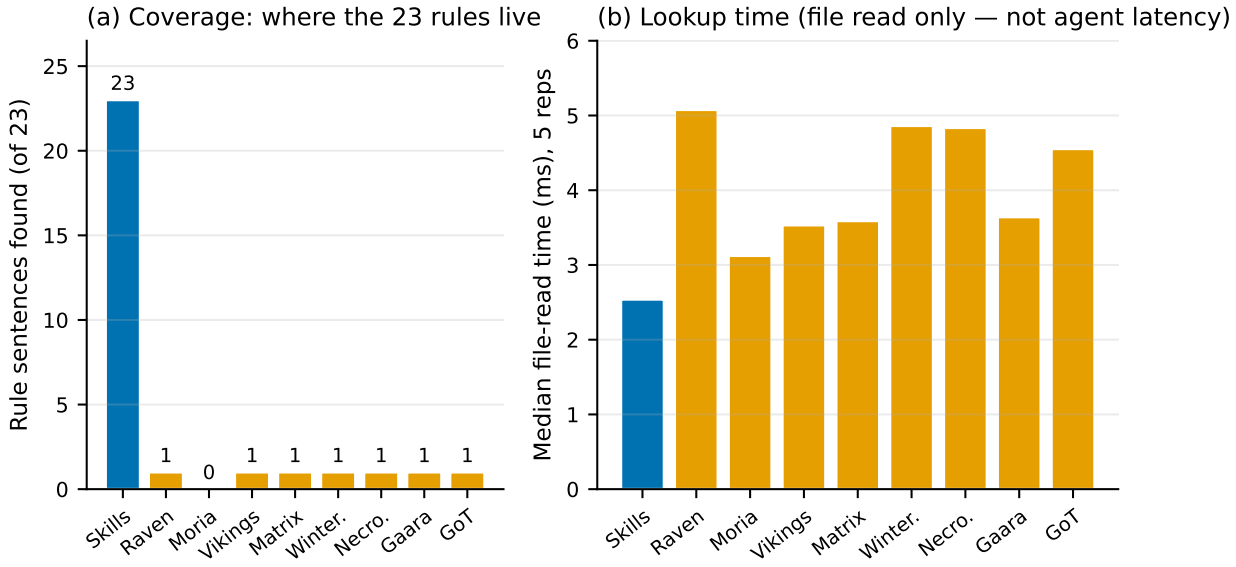



Figure 10: Experiment A. (a) Coverage is the result: the rules live in the skills (23/23) and not in finished lab files (0–1/23). (b) Median file-read times differ by milliseconds and carry no agent meaning; they are shown to preclude, not support, a latency reading.

The timing columns require the paper's most explicit non-claim. All corpora are read in single-digit milliseconds (skills median 2.539 ms; lab medians 3.123–5.076 ms in Experiment A; 0.562–4.284 ms across Experiment B), repetition 1 is the cold read (e.g., 8.842 ms for the skills corpus in Experiment A), and the ordering between corpora is not even consistent across experiments. These are measurements of a script that already knows the sentence it is looking for. They are not LM task times, they do not show that an agent with the skills is faster, and we draw no speed conclusion from them. The qualitative speed-adjacent observation the sessions do support is avoided iteration: before codification, the Raven conversion spent rebuilds on the wrong suite and engine and probed Sendmail inside its startup minute; the codified rules remove those specific iterations from the next conversion by stating the base, engine, flag, and wait in advance. Testing whether that removal materialises as agent time saved requires timing a second agent, which was not done.

## 5. Discussion

### *5.1. What the results mean*

Three findings survive the paper's own caveats. First, *container replicas of owned appliances can be*

Table 9: Experiment A: exact-string lookup of 23 rule sentences (13 Oct 2026, five repetitions, median of five). Times are script file-read times. Experiment B per-lab medians (skills vs. replica files) were: Gaara 1.900/3.461 ms, Game of Thrones 1.874/4.284 ms, Matrix 0.593/0.562 ms, Necromancer 1.724/4.127 ms, Vikings 0.618/0.635 ms, Wintermute 0.758/1.039 ms (MrRobot, sensitivity: 2.129/4.050 ms); coverage was 23/23 vs. 1/23 in every run.

| Corpus | Files | Hits | Median (ms) |
|---|---|---|---|
| Three skills | 3 | 23/23 | 2.539 |
| Raven files | 8 | 1/23 | 5.076 |
| Moria files | 5 | 0/23 | 3.123 |
| Vikings files | 6 | 1/23 | 3.532 |
| Matrix files | 6 | 1/23 | 3.589 |
| Wintermute files | 9 | 1/23 | 4.862 |
| Necromancer files | 9 | 1/23 | 4.835 |
| Gaara files | 7 | 1/23 | 3.641 |
| Game of Thrones files | 9 | 1/23 | 4.553 |

*made configuration-faithful, but fidelity is a scored property, not a by-product of a working build.* Every replica in the corpus answered on its published address, yet the strict pass rate is 83.0%, not 100%, and the shortfall is concentrated exactly where casual verification does not look: package revision strings and the full listener set. A "does it respond?" acceptance test—the analogue of accepting an agent's plausible narrative (Jimenez et al., 2024)—would have reported eight successes and hidden fifteen substitutions.

Second, *the fidelity frontier is archival, not structural.* Identity, privilege, containment, and application state transferred perfectly (32 of 32 rows across users, sudoers, loopback, and URL); what resisted transfer is the past itself. EOL archives serve newer revisions than guests carried, non-Linux guests have no container base, and guest-runtime state (a Docker engine inside the guest, systemd-resolved's ephemeral listener) has no file to copy. This reframes VM-to-container conversion as a problem in *software archaeology under trust constraints*: the snapshot-pinning and vault-mirror mechanisms that recovered the only two package-row passes (Raven, Moria) are the same mechanisms Caneill et al. (2017) and Lamb & Zacchiroli (2022) motivate for historical and reproducible builds, and the rubric's refusal to call a newer revision a pass is what makes their use auditable rather than cosmetic.

Third, *the durable contribution of an agent session is the procedure it leaves behind.* Raven's six fixed rows were not fixed by a better single build; they were fixed by converting session discoveries into preconditioned rules with stop conditions, after which the rescore moved. RQ4 sharpens the point negatively: the finished files retain one of 23 rules, so an agent (or a human) starting from the artifacts alone inherits the answers without the criteria that distinguish a pass from a confident partial. That is a general hazard for LM-assisted systems work as its outputs accumulate: *artifacts under-specify the evaluations that legitimised them*, and evaluations therefore need to be published as first-class, re-runnable artifacts alongside the builds they score—here, the skills, the inventories, and the parity reports.

### *5.2. Security implications*

The parity discipline is also a containment discipline. The loopback-unpublished row passing on all eight labs (plus MrRobot) means no replica silently widened its guest's attack surface: databases, mail submission, Redis, SNMP, and admin listeners that the guests kept private stayed private, while published surfaces kept the guest's own port numbers, preserving the enumeration lesson the lab exists to teach (Thompson & Irvine, 2018; Yamin et al., 2020). Conversely, the rubric treats widening as the severest outcome (a missing row), which inverts the usual convenience gradient under which exposing a database "temporarily, for debugging" is costless. The EOL trust rules carry the second security implication: `trusted=yes` and disabled validity checking are necessary to install EOL guests at all, and are therefore precisely the settings that must be line-scoped, failure-triggered, and permanently registered as gaps, lest a repair for a 2015 suite become a standing policy for current mirrors. Finally, nothing in the conversion requires attacking the guest: administration is recovered offline on a clone, inventory stops at version recording, and reports are secret-free by construction, which keeps the workflow inside the authorisation boundary of owned appliances. The student surface strengthens the same properties—current base, no root SSH, learner-chosen credentials—so the teaching artifact does not re-distribute the historical vulnerable system it describes. None of this makes a replica *safe* in the container-security sense: replicas intentionally run old, vulnerable services and inherit the shared-kernel exposure Sultan et al. (2019) catalogue; they belong on isolated

laboratory networks, which the dedicated-address publication model is designed to enforce.

### *5.3. Threats to validity*

*Construct validity..* Parity is configuration equality against an inventory, not behavioural equivalence: we did not fuzz services, replay workloads, or verify that an exercise's full solution path survives conversion, and a pass row for "web paths" evidences presence and HTTP status, not byte-identical application behaviour in all cases (Game of Thrones' uncopied web trees are scored partial for this reason). Memory results are configured caps, not residency, latency, or throughput measurements. The lookup study measures textual residence of rules, not agent comprehension or task success.

*Internal validity..* Row scoring was performed by the project that built the replicas, against its own rubric, without independent re-scoring; the Raven before/after comparison entangles rule codification with the rebuild itself and with scorer learning. Five of eight inventories rest on configuration evidence from unbooted guests, so their listener rows could err in either direction (a configured service that would not have started; a started service with no surviving configuration). The exploratory correlations use eight heterogeneous cases and are reported without inferential claims.

*External validity..* The corpus is the project's owned appliances: small, old, Linux/BSD, web-and-SSH-centric boot-to-root systems on one Windows/WSL2 + VirtualBox host. Results should not be extrapolated to Windows guests, current distributions, cloud ranges, multi-host enterprise scenarios (Chouliaras et al., 2023), or conversions performed by other agents or humans. The one population the results do speak to—educators and range operators holding similar legacy appliances—is the population the skills address.

*Reliability..* The audit script re-derives headline counts from the primary files and is retained with the paper materials, as are the sentence set, the per-lab experiment files, and the figure-generation script from the same dataset table that the audit validated. Independent replication requires the owned appliances (not redistributable) but not the sessions: skills, inventories, Dockerfiles, Compose files, and parity reports suffice to re-score every row except those requiring a booted guest, whose provenance is marked.

### *5.4. Limitations and future work*

The decisive unrun experiment is a *second-agent study*: give an LM agent that has not seen the sessions a new owned appliance, with and without the skills, and measure parity outcome and task time under the same rubric. Until that is run, RQ3–RQ4 support a residence and mechanism claim, not an efficacy or speedup claim. Three engineering evaluations are similarly open: measured (rather than configured) memory, startup, and throughput of replicas against their guests, in the manner of Felter et al. (2015) and Nakata & Otsuka (2021); behavioural exercise-level validation (can each lab's documented learning path be completed against the replica, and does automated assessment agree with the VM's, as in Labtainers-style grading (Thompson & Irvine, 2018)); and independent re-scoring of the 88 rows to estimate rubric reliability. On the method side, the rubric should grow rows the current corpus did not force—kernel-adjacent behaviour, filesystem state beyond web trees, and timing behaviour—and the scenario-design skill's outputs should be evaluated pedagogically (learning outcomes, not only topology well-formedness) before classroom claims are made. The corrected README sentence (audit item 1) should be fixed upstream so the narrative record again matches the scored artifacts.

## 6. Conclusion

An owned virtual appliance and a container that answers on the same ports are not automatically the same laboratory. This paper supplied the missing middle: a six-skill conversion procedure with stop conditions, an eleven-row parity rubric that scores configuration equality from live evidence, and a gap register that keeps archive trust, the shared host kernel, and supervisor-for-init permanently visible instead of letting them be scored away. Across eight replicas (88 rows), the procedure yields 73 passes, 15 named partials, and no missing rows, inside configured memory envelopes totalling 35.7% less than the guests' RAM allocations; package-revision archaeology and listener completeness, not identity, privilege, containment, or application state, are where fidelity is won and lost. Codifying the

procedure moved the one longitudinally scored replica from five to eleven passing rows, and the rules that did so demonstrably live in the skill documents rather than in the finished builds (23/23 vs. 7/184 lab-file hits). The claims are deliberately bounded—no measured residency, no second-agent timing, no kernel equivalence— because the paper's central argument is that bounded, checkable claims are the product. For educators and range operators holding legacy appliances, the artifact set (skills, redacted inventories, pinned builds, parity reports, and teaching modules) offers a way to retire the hypervisor seat without retiring the host the lesson was about—and to prove, row by row, what survived the move.


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## CRediT authorship contribution statement

**ZeroCrowd-Labs Project Team:** Conceptualisation, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing—original draft, Writing—review and editing, Visualisation. Individual contributor names and roles to be completed on submission.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The conversion skills, redacted guest inventories, Dockerfiles, Compose files, parity reports, lookup sentence set, validation script, and figure-generation code are available in the ZeroCrowd-Labs repository accompanying this manuscript. Guest disk images, appliance archives, credentials, and shadow material are not shared: they are neither necessary to re-score the reported rows nor redistributable.

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